Variable Stars in the VO

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Lightcurves of variable stars

Use cases:

- data for supervised machine learning
- long-term behaviour (period changes)
- historical outbursts of transients

Where are lightcurves?

- Large surveys
- Small archives spread across:
  - personal disks
  - institutional archives,
  - sometimes only as figures in papers,
  - sometimes in VizieR,
  - sometimes somewhere in the internet

These data should be discoverable and accessible

We have published:

- our timeseries and images from Kolonica observatory
- personal collections (first “seed” datasets from individual observers)
- OGLE Collection of Variable Stars (re-published in VO-compliant format)

OGLE Collection of Variable Stars

Optical Gravitational Lensing Experiment
running since 1992

~2 billion measurements in I and V bands
>1 million variable stars
shipped in about 200 files (without light curves)
https://www.astrouw.edu.pl/ogle/ogle4/OCVS/

Now is accessible via TAP, Cone Search, SSA, ObsCore
with previews of folded and unfolded lightcurves



